

Ques.1. Print numbers from 1 to 10. If a number is even, break the loop using a for loop and else clause.

```
for n in range(1, 11):
    print(n)
    if n % 2 == 0:
        print("Even number found, breaking the loop. the even number is", n+1)
        break
else:
    print("Loop completed without a break.")
```

```
1
2
Even number found, breaking the loop. the even number is 3
```

```
num = 1
while num <=10:
    print(num, end = " ")
    num +=1
```

```
1 2 3 4 5 6 7 8 9 10
```

Q2, Print numbers from 1 to 5, except 3 using a for loop and continue statement

```
for n in range (1,10):
    if n==3:
        continue
    print(n)
```

```
1
2
4
5
6
7
8
9
```

Q3. Print numbers in a list until a negative number is encountered using a while loop.

```
numbers = [1,2,3,0,-1,-2,-3]
n=0

while n < len(numbers) and numbers[n]>=0:
    print(numbers[n])
    n += 1
```

```
1
2
3
0
```

Q4. Finding Common elements in two lists using for loop List=[1,2,3,4,5] List = [3,4,5,6,7]

```
list1 = [1,2,3,4,5]
list2 =[3,4,5,6,7]
ce= []
for item in list1:
    if item in list2:
        ce.append(item)
    print("Common Elements:", ce)
```

```
Common Elements: [3]
Common Elements: [3, 4]
Common Elements: [3, 4, 5]
```

##Q5. Print the Fibonacci sequence up to the 10th term using a while loop

```
terms=10
n1,n2 = 0,1
count = 0

print("Fibonacci Sequence:")
while count< terms:
    print(n1,end=" ")
    nth = n1+n2
    n1 = n2
    n2=nth
```



```
count +=1
print ()
```

```
Fibonacci Sequence:
0
1
1
2
3
5
8
13
21
34
```

```
## 6. Print numbers divisible by 3 or 5 from 1 to 20 using a for loop.
```

```
for number in range (1,21):
    if number % 3 == 0 or number % 5 == 0:
        print(number)
```

```
3
5
6
9
10
12
15
18
20
```



```
## 7. Find all prime numbers between 1 and 50 using nested for and if.
```

```
prime = []
for number in range (2,51):
    for i in range (2,number):
        if number % i == 0:
            break
    else:
        prime.append(number)

print ("Prime numbers between 1 and 50 are :", prime)
```

```
Prime numbers between 1 and 50 are : [2]
Prime numbers between 1 and 50 are : [2, 3]
Prime numbers between 1 and 50 are : [2, 3, 5]
Prime numbers between 1 and 50 are : [2, 3, 5, 7]
Prime numbers between 1 and 50 are : [2, 3, 5, 7, 11]
Prime numbers between 1 and 50 are : [2, 3, 5, 7, 11, 13]
Prime numbers between 1 and 50 are : [2, 3, 5, 7, 11, 13, 17]
Prime numbers between 1 and 50 are : [2, 3, 5, 7, 11, 13, 17, 19]
Prime numbers between 1 and 50 are : [2, 3, 5, 7, 11, 13, 17, 19, 23]
Prime numbers between 1 and 50 are : [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]
Prime numbers between 1 and 50 are : [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31]
Prime numbers between 1 and 50 are : [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37]
Prime numbers between 1 and 50 are : [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41]
Prime numbers between 1 and 50 are : [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43]
Prime numbers between 1 and 50 are : [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]
```

```
## Q8. Find the first occurrence of a number in a list using a while loop.
```

```
def find_first_occurrence_while(data_list, target_number):
    index = 0
    while index < len(data_list):
        if data_list[index] == target_number:
            return index
        index += 1
    return -1
```

```
my_list = [10, 20, 30, 20, 40, 50,0]
number_to_find = 0
first_index = find_first_occurrence_while(my_list, number_to_find)
```

```
if first_index != -1:
    print(f"The first occurrence of {number_to_find} is at index {first_index}.")
else:
    print(f"{number_to_find} not found in the list.")
```

```
number_to_find_not_present = 1
first_index_not_present = find_first_occurrence_while(my_list, number_to_find_not_present)
```

```
if first_index_not_present != -1:
    print(f"The first occurrence of {number_to_find_not_present} is at index {first_index_not_present}.")
```

```

else:
    print(f"{number_to_find_not_present} not found in the list.")

```

The first occurrence of 0 is at index 6.
1 not found in the list.

Q9. Calculate the factorial of a number using a while loop.

```

def calculate_factorial(n):
    if n < 0:
        return "Factorial is not there for a negative number"
    elif n == 0:
        return 1
    else:
        factorial = 1
        current_number = n
        while current_number > 0:
            factorial *= current_number
            current_number -= 1
        return factorial

```

```

number = 3
result = calculate_factorial(number)
print(f"The factorial of {number} is: {result}")

```

```

number_negative = -2
result_negative = calculate_factorial(number_negative)
print(f"The factorial of {number_negative} is: {result_negative}")

```

```

number_zero = 0
result_zero = calculate_factorial(number_zero)
print(f"The factorial of {number_zero} is: {result_zero}")

```

The factorial of 3 is: 6
The factorial of -2 is: Factorial is not there for a negative number
The factorial of 0 is: 1

Q.10 Print pattern using nested loop

```

rows = 10
for i in range(1, rows+1):
    for u in range(i):
        print("*", end=" ")
    print()

```

```

*
* *
* * *
* * * *
* * * * *
* * * * * *
* * * * * * *
* * * * * * *
* * * * * * *
* * * * * * *

```

```

rows = 10

```

```

for i in range(rows, 0, -1):
    for u in range(i):
        print("*", end=" ")
    print()

```

```

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
* * * *
* * * *
* * *
* *
*

```

```

xrows = 5

```

```

for i in range(1, rows + 1):
    print(" " * (rows - i) + "*" * (2 * i - 1))

```

```

*
***

```

```
*****
*****
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*****
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*****
*****
```

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